

of hardware and software. The tools can be categorised under network analysis, reverse engineering and debugging, Web based analytics, forensics applications, malware analysis and packets investigation. A few of the tools under each category are described below.

Network analysis: The tools for network analysis are Dirlbags py-pcap, dpkt, flowgrep, Habu, Impacket, etc.

Reverse engineering and debugging: Androguard, CHIPSEC, Capstone, Frida, IDAPython, etc, may be used for reverse engineering and debugging.

Web based analytics: Some of the Web based analytics tools are FunkLoad, Ghost.py, HTTPie, ProxMon, Requests, etc.

Forensics applications: Aft, LibForensics, Rekall, Volatility, TrIDLib, etc, can be used for forensics applications.

Malware analysis and packets investigation: For malware analysis and packets investigation, CapTipper, Exefilter, jsunpack-n, phoneyc, PyClamAV, etc, may be used.

The PytheM penetration testing framework

A large set of software tools and libraries written in Python is available for simulation of different types of attacks. Python is rich in the additional plugins and modules that can be attached for high performance forensics applications and cyber security. PytheM is one of the powerful tools which uses Python at the back-end. It provides functions to test the network and Web applications for different types of attacks before actual deployment.

PytheM is a free and open source penetration testing framework with multi-functional features to analyse the network and Web deployment to prevent multiple attacks. It assists security professionals and administrators in evaluating and performing security audits of their infrastructure. You can find more on PytheM at <https://github.com/m4n3dw0lf/pythem>.

Downloading and installation instructions

PytheM can be installed without any complexity on Linux/GNU platforms. It can be installed and executed on Docker containers.

Given below is the code to install PytheM on Ubuntu systems:

```
$ sudo apt-get update
$ sudo apt-get install -y build-essential python-dev python-
pip tcpdump python-capstone libnetfilter-queue-dev libffi-dev
libssl-dev
```

To install PytheM using Pip, use the following command:

```
$ sudo pip install pythem
```

The code for installing PytheM using the Git repository is:

```
$ git clone https://github.com/m4n3dw0lf/pythem
```

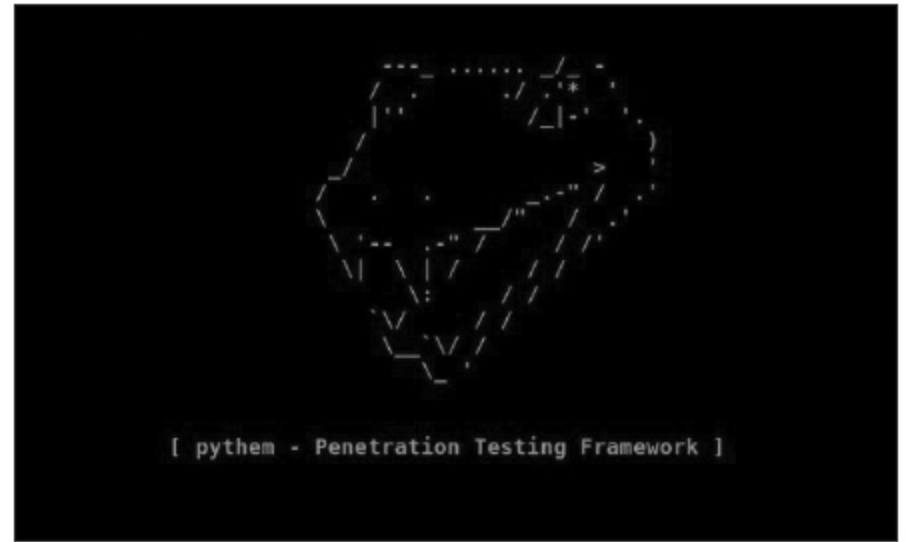


Figure 1: The PytheM penetration testing framework

```
$ cd pythem
$ sudo python setup.py install
```

To install PytheM by integrating the source and Pip, use the following code:

```
$ git clone https://github.com/m4n3dw0lf/pythem
$ cd pythem
$ sudo python setup.py sdist
$ sudo pip install dist/*
```

To execute and run PytheM with root privileges, type:

```
$ sudo pythem
```

To install PytheM on Docker, give the following command:

```
docker run -it --net=host --rm --name pythem m4n3dw0lf/pythem
```

Analysis of different attacks using PytheM

There are many types of attacks in the network environment, which can be avoided if tested by network administrators. It is the responsibility of network and Web administrators to evaluate their environment for different cracking attempts so that a secured mechanism can be launched.

PytheM provides the features to implement different types of attacks for penetration testing of the network deployment. Using these assaults, security professionals can predict the vulnerabilities in their network.

The following are a few of the attacks that can be simulated using PytheM:

- Man-in-the-middle attacks
- ARP spoofing
- DHCP spoofing
- Brute force attacks
- ACK injection
- PCAP analysis
- URL busters
- Overthrow DNS
- Redirections, and many others